

NORTHEAST ALABAMA, AL, USA

WMO: 722285

Lat: 33.967N

Lon: 86.083W

Elev: 173

StdP: 99.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.2	-14.8	1.1	-4.5	-12.3	1.3	-2.0	8.6	7.0	7.8	6.9	1.6	340	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.5	23.9	33.1	23.8	32.4	23.7	26.0	31.2	25.5	30.8	24.9	30.1	2.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.2	28.4	23.9	19.2	27.9	23.1	18.2	27.3	81.6	31.2	78.8	30.7	76.3	30.7	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.4	6.4	5.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-10.9	35.8	2.7	2.1	-12.8	37.3	-14.4	38.5	-15.9	39.6	-17.9	41.1
			-11.5	26.7	2.4	0.8	-13.2	27.3	-14.6	27.8	-15.9	28.3	-17.7	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	5.3	8.1	11.9	15.8	20.4	24.4	26.2	25.8	22.6	16.5	10.2	6.6
	DBStd	8.40	5.55	5.16	4.90	4.01	3.37	2.20	1.82	2.01	3.30	4.45	4.55	4.96
	HDD10.0	486	167	88	38	5	0	0	0	0	5	52	132	
	HDD18.3	1714	403	287	203	95	21	1	0	7	88	245	365	
	CDD10.0	2747	22	36	97	178	323	432	501	491	378	205	59	26
	CDD18.3	934	1	1	5	18	85	182	243	233	134	30	2	1
	CDH23.3	8550	1	7	67	242	826	1636	2218	2091	1152	292	18	1
	CDH26.7	3503	0	0	7	36	262	702	1011	964	451	69	2	0
(14) Wind	WSAvg	1.9	2.4	2.4	2.5	2.2	1.8	1.6	1.5	1.4	1.7	1.8	1.9	2.1
(15) Precipitation	PrecAvg	1401	133	135	148	121	106	114	129	102	97	74	110	131
	PrecMax	1882	220	248	237	228	337	272	246	220	224	250	267	250
	PrecMin	745	58	64	46	38	16	53	33	14	15	2	33	54
	PrecStd	232	47	46	45	48	65	56	55	43	62	57	51	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	24.2	27.6	29.1	32.5	35.1	36.2	37.2	34.8	31.1	26.2	22.3
		MCWB	16.0	17.1	17.9	19.0	22.7	23.4	24.3	24.1	22.8	21.9	18.4	16.6
	2%	DB	19.0	21.7	25.1	27.7	31.1	33.0	34.1	34.5	32.9	28.3	22.8	19.3
		MCWB	14.9	16.3	16.9	18.6	22.1	23.2	24.3	24.2	22.7	20.3	16.4	15.8
	5%	DB	17.2	19.2	23.0	26.3	29.1	32.3	32.9	32.9	31.2	27.0	21.2	17.5
		MCWB	13.6	14.8	15.5	17.7	21.3	23.4	24.4	24.0	22.4	18.7	15.6	14.1
	10%	DB	14.8	17.4	21.2	24.2	27.9	31.0	32.1	32.0	29.1	24.3	18.8	15.2
		MCWB	11.7	12.7	14.7	16.9	20.7	23.1	24.4	23.8	22.0	18.2	14.5	12.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	19.6	19.6	21.4	24.6	26.5	26.9	26.7	25.4	24.2	20.4	18.7
		MCDB	20.1	22.0	24.3	26.5	29.0	30.7	31.5	32.1	30.2	28.0	22.5	20.8
	2%	WB	15.9	17.9	18.5	20.1	23.5	25.3	26.2	25.8	24.5	22.5	18.6	16.5
		MCDB	18.2	20.3	23.0	25.3	28.6	30.7	31.2	31.4	29.4	26.0	22.0	18.7
	5%	WB	14.4	15.8	17.2	19.1	22.6	24.6	25.6	25.3	23.7	20.5	16.9	14.8
		MCDB	16.7	18.3	21.3	24.3	27.7	29.8	30.8	30.9	28.6	24.5	19.7	17.4
	10%	WB	12.0	13.8	15.8	18.0	21.7	23.9	25.0	24.6	22.9	19.2	14.9	12.6
		MCDB	14.3	16.2	20.0	22.7	26.6	28.7	30.2	29.9	27.6	23.4	18.1	14.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.2	13.1	14.1	13.0	11.7	10.9	11.4	12.2	13.6	13.7	11.4
		MCDBR	13.5	14.9	16.0	16.0	15.0	14.3	12.6	13.6	14.3	15.6	16.0	13.7
	5% WB	MCWBR	10.0	10.0	9.1	8.4	7.2	6.0	5.0	5.3	5.8	8.1	10.3	10.4
		MCDWR	11.0	11.9	13.0	13.2	12.4	11.7	11.3	11.6	11.5	11.9	12.2	11.2
(40) Clear-Sky Solar Irradiance	taub	0.320	0.323	0.363	0.395	0.461	0.486	0.535	0.536	0.434	0.376	0.343	0.322	
		2.506	2.493	2.404	2.299	2.156	2.127	2.007	1.992	2.271	2.418	2.450	2.515	
	Ebn at Noon	893	928	912	892	833	809	767	758	828	855	854	869	
		84	95	112	130	152	156	175	175	127	101	89	79	
(44) All-Sky Solar Radiation	RadAvg	2.57	3.09	4.23	5.46	5.96	6.16	5.91	5.48	4.77	3.96	2.91	2.19	
	RadStd	0.18	0.40	0.39	0.37	0.48	0.48	0.38	0.30	0.43	0.51	0.32	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	+113

Nomenclature: See separate page